Journal of Manufacturing Systems 70 (2023) 137–148

Contents lists available at ScienceDirect

Journal of Manufacturing Systems

journal homepage: www.elsevier.com/locate/jmansys

Energy consumption prediction and optimization of industrial robots based on LSTM

Pei Jiang [a], Zuoxue Wang [a], Xiaobin Li [a,*], Xi Vincent Wang [b], Bodong Yang [a], Jiajun Zheng [a]

[a] State Key Laboratory of Mechanical Transmissions, Chongqing University, Chongqing, 400044, China
[b] Department of Production Engineering, KTH Royal Institute of Technology, Stockholm, SE-10044, Sweden

ARTICLE INFO

Keywords:
Industrial robots
Data-driven
LSTM
Time scaling
Energy optimization

ABSTRACT

Due to wide distribution and low energy efficiency, the energy-saving of industrial robots draws more and more attention, and a large number of methods have emerged to predict or optimize the energy consumption (EC) of robots. However, many dynamic and electrical parameters are unavailable due to the commercial limitations of industrial robots, which constrains the application of those model-based methods. Therefore, this paper proposes a data-driven method for the prediction and optimization of robot EC. Initially, the cause-and-effect relationship between robot EC and joint motion variables, such as the joint position, velocity, and acceleration, is qualitatively analyzed based on the influence of the capacitive and inductive components in the drive system. And a deep neural network based on long short-term memory (LSTM) is proposed to reveal the nonlinear mapping between the industrial robot EC and the joint motion variables, which can predict EC without the parameters of the industrial robot. Based on the proposed neural network, the adaptive genetic algorithm is adopted to optimize the time-variant scaling function, which can optimize the scaled trajectory to reduce EC without hardware modification. To validate the accuracy and efficacy of the proposed method, experiments are conducted on a KUKA KR60-3 six degree-of-freedom (DOF) industrial robot. The results demonstrate that the proposed neural network can predict EC with a mean absolute percentage error less than 4.21% and the proposed method reduces the EC by 22.35%.

1. Introduction

Nowadays, industrial robots have been widely used in handling, assembly, welding, grinding, exploration and other industries due to their advantages of precise movement, high reliability, high load and programming flexibility [1–5]. 3.5 million industrial robots has being used in factories worldwide by 2022 [6]. However, the EC of industrial robots is featured with high EC and low energy efficiency. Researches reveal that the EC cost of robots accounts for about 60% of the total follow-up cost [7], while the most of the energy is consumed by the iron loss, copper loss, joint friction, etc. Due to the low energy efficiency and wide distribution, the EC optimization of industrial robots is of great significance for the sustainable development of the manufacturing industry [8–10].

To reduce the EC of industrial robots, researchers mainly focus on eco-efficient design and planning [11]. The former mainly reduces EC by lightweight design, introducing new energy-saving hardware or adjusting the layout of the production line [12–16]. However, the adjustment of hardware requires considerable investment, which limits its application in practical application. The latter mainly focuses on the EC optimization of industrial robots by trajectory planning and production scheduling [17–19].

For trajectory planning, an accurate description of the EC of industrial robots is the basis of trajectory optimization. Hansen et al. took into account the friction loss and the loss of servo drivers and inverters [20]. Pellicciari et al. considered the dynamic coupling behavior between permanent magnet synchronous motors, servo controllers, inverters and mechanical systems, and a virtual prototype-based EC model is developed [21]. Brossog et al. analyzed the effects of speed, load and trajectory smoothness on the robot EC [22]. Based on the analysis of energy exchange on the DC bus capacitor, Meike et al. proposed a robot energy conversion criterion to reduce EC [23]. Furthermore, the energy characteristics of robotic manufacturing systems have also been analyzed [24–27]. However, since constructing an accurate EC model requires considering many parameters and dynamic processes in the circuit, the implementation in the industrial environment is too cumbersome. Thus, the above research has simplified the EC calculation of the drive circuit system to a certain extent to complete the construction of the EC model.

* Corresponding author.
E-mail address: xiaobin_lee@cqu.edu.cn (X. Li).

https://doi.org/10.1016/j.jmsy.2023.07.009
Received 1 December 2022; Received in revised form 26 June 2023; Accepted 18 July 2023
Available online 28 July 2023
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Nomenclature

α	Feed-rate override
$\boldsymbol{b}_{\mathrm{f}}$, $\boldsymbol{b}_{\mathrm{i}}$, $\boldsymbol{b}_{\mathrm{c}}$, $\boldsymbol{b}_{\mathrm{o}}$	Bias terms corresponding to forget gate, input gate, state of LSTM cell and output gate, respectively
$\boldsymbol{c}_{t(k)}$	State of LSTM cell at time t_{k}
$\boldsymbol{f}_{t(k)}$, $\boldsymbol{i}_{t(k)}$, $\boldsymbol{h}_{t(k)}$, $\boldsymbol{o}_{t(k)}$	Output terms of forget gate, input gate, LSTM cell and output gate at time t_{k}, respectively
$\boldsymbol{W}_{\mathrm{xc}}$, $\boldsymbol{W}_{\mathrm{hc}}$	Weight coefficient matrices corresponding to state of LSTM cell
$\boldsymbol{W}_{\mathrm{xf}}$, $\boldsymbol{W}_{\mathrm{hf}}$	Weight coefficient matrices of forget gate
$\boldsymbol{W}_{\mathrm{xi}}$, $\boldsymbol{W}_{\mathrm{hi}}$	Weight coefficient matrices of input gate
$\boldsymbol{W}_{\mathrm{xo}}$, $\boldsymbol{W}_{\mathrm{ho}}$	Weight coefficient matrices of output gate
$\boldsymbol{\tau}$	Joint torque vector of robot, $\boldsymbol{\tau} \in \mathbb{R}^n$
$\boldsymbol{\tau}_{\mathrm{e}}$	Equivalent external torque vector of robot, $\boldsymbol{\tau}_{\mathrm{e}} \in \mathbb{R}^n$
$\boldsymbol{C}$	Coriolis and centrifugal torque, $\boldsymbol{C} \in \mathbb{R}^{n\times n}$
$\boldsymbol{f}$	Equivalent friction torque of robot, $\boldsymbol{f} \in \mathbb{R}^n$
$\boldsymbol{G}$	Gravity torque, $\boldsymbol{G} \in \mathbb{R}^n$
$\boldsymbol{M}$	Inertia matrix of robot, $\boldsymbol{M} \in \mathbb{R}^{n\times n}$
$\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\ddot{\boldsymbol{q}}$	Joint position, velocity and acceleration vectors, respectively
$\boldsymbol{q}_{\mathrm{r}}(t_{\mathrm{r}})$, $\dot{\boldsymbol{q}}_{\mathrm{r}}(t_{\mathrm{r}})$, $\ddot{\boldsymbol{q}}_{\mathrm{r}}(t_{\mathrm{r}})$	Reference position, velocity and acceleration trajectories of robot joint, respectively
$\boldsymbol{q}_{\mathrm{s}}(t)$, $\dot{\boldsymbol{q}}_{\mathrm{s}}(t)$, $\ddot{\boldsymbol{q}}_{\mathrm{s}}(t)$	Scaled position, velocity and acceleration trajectories of robot joint, respectively
$\boldsymbol{q}_{\mathrm{r},k}$, $\dot{\boldsymbol{q}}_{\mathrm{r},k}$, $\ddot{\boldsymbol{q}}_{\mathrm{r},k}$	Values of reference joint position, velocity, acceleration at the moment $t_{\mathrm{r}} = t_{\mathrm{r},k}$, $k \in \{0, 1, \ldots, N\}$, respectively
$\boldsymbol{q}_{\mathrm{s},k}, \dot{\boldsymbol{q}}_{\mathrm{s},k}, \ddot{\boldsymbol{q}}_{\mathrm{s},k}$	Values of scaled joint position, velocity, acceleration at the moment $t_{\mathrm{r}} = t_k$, $k \in \{0, 1, \ldots, N\}$, respectively
Δ	Sampling interval at reference time frame
$\dot{\boldsymbol{q}}_{\max}$, $\ddot{\boldsymbol{q}}_{\max}$	Thresholds of joint velocity and acceleration, respectively
$\dot{u}, \ddot{u}$	First order and second order derivatives of u with respect to t
σ	Sigmoid activation function
φ_{ac}	Nonlinear mapping from $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\ddot{\boldsymbol{q}}$ to P_{ac}
φ_{add}	Nonlinear mapping from P_{pmsm} to P_{add}
φ_{Cu}	Nonlinear mapping from $\boldsymbol{\tau}$ to P_{Cu}
φ_{Fe}	Nonlinear mapping from $\dot{\boldsymbol{q}}$ to P_{Fe}
φ_{mec}	Nonlinear mapping from $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\ddot{\boldsymbol{q}}$ to P_{mec}
φ_{pmsm}	Nonlinear mapping from $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\ddot{\boldsymbol{q}}$ to P_{pmsm}
A	Adjustable hyper-parameter of AGA
E_{ac}	Total energy consumption of the robot
fit_{avg}	Average fitness level of population
fit_{c}	Average fitness level of two crossover individual
fit_{i}	Fitness level of individual
$fit_{\max}$	Maximum fitness level of population
i_{pmsm}	Equivalent current of PMSMs, $i_{\mathrm{pmsm}} \in \mathbb{R}^n$
K_{gea}	Gear ratios, $K_{\mathrm{gea}} \in \mathbb{R}^{n\times n}$
K_{invc}	Conduction coefficient matrix, $K_{\mathrm{invc}} \in \mathbb{R}^{n\times n}$
k_{invsw}	Switching losses coefficient, $k_{\mathrm{invsw}} \in \mathbb{R}^n$
K_{tor}	Motor torque constants, $K_{\mathrm{tor}} \in \mathbb{R}^{n\times n}$
n	Number of joint
N_{u}, N_{v}, N_{t}	Dynamic programming grid parameters
P_{ac}	Total power of the robot
P_{add}	Miscellaneous losses and possible calculation error
P_{Cu}, P_{Fe},	Copper loss and iron loss, respectively
P_{c}, P_{cmax}, P_{cmin}	Individual crossover probability and its upper and lower bounds, respectively
P_{dc}, P_{cap}, P_{inv}, P_{pmsm}, P_{mec}	Input power of the DC bus, the DC bus capacitance, the inverters, PMSMs and the mechanical system, respectively
P_{m}, P_{mmax}, P_{mmin}	Individual mutation probability and its upper and lower bounds, respectively
P_{other}	Other energy loss which is mainly original from the cooling fans, the PC in the control cabinet and the control panel
P_{reclos}, P_{invlos}, P_{rd}	Power loss on the rectifier, inverters and the brake resistance, respectively
$t_{\max}$	Thresholds of execution time
t_{r}, t	Reference and scaled time frame, respectively
tanh	Hyperbolic tangent activation function
$u(\cdot)$	Time scaling function, $u(\cdot) \in \mathbb{R}^+$
v_{dc}, $v_{\max}$, v_{sta}	DC bus voltage and its maximum and standard values, respectively
h_{k}	Time interval after scaling
u_{k}	Values of time scaling function at the moment $t_{\mathrm{r}} = t_{\mathrm{r},k}$, $k \in \{0, 1, \ldots, N\}$
AC	Alternating current
AGA	Adaptive genetic algorithm
BP	Back propagation neural network
DC	Direct current
DOF	Degree of freedom
DPBEPE	Dynamic programming based on energy parameterization equation
ECPE	Energy consumption parameterization equation
EC	Energy consumption
GRU	Gate Recurrent Unit
LSTM	Long-short time memory
MAE	Mean absolute error

Based on the EC model, a large amount of trajectory optimization methods are proposed [28–32]. Among those various optimization methods, time scaling technology has drawn increasing attention due to its ability to optimize trajectories in the time domain without altering the motion path in the spatial domain. Pellicciari et al. formulated the EC of an industrial robot system as a function of a linear scaling factor

MAPE	Mean absolute percentage error
MSE	Mean square error
NN	Neural network
PMSM	Permanent magnet synchronous machine
PTP	Point-to-point
PWM	Pulse width modulation wave
RNN	Recurrent neural network

and optimized the EC of the pick-and-place task [31]. Meike et al. investigated the role of linear time scaling technique in optimizing the EC of multi-robot systems in automotive automated assembly lines while ensuring productivity [10]. Vergnano et al. also used linear scaling method to optimize the EC of a multi-robot scheduling problem [28]. In [32], Wigstrom et al. adopted the dynamic time scaling mapping to scale the trajectory, and the dynamic programming algorithm was also applied to achieve optimal EC. Li et al. developed an energy characteristic parameterization model based on dynamic time scaling, which can replace the integral operation of transient power with algebraic operation [33], improving the computation efficiency of dynamic programming and reducing the computation time by nearly 99%. Since the time scaling technology only reduces the EC by changing the trajectory of the robot in the time domain, it has the characteristics of simple and fast operation, allowing the time function to change independently, and selecting more types of scaling functions [29]. By applying this technology to reduce the EC of the robot, no additional investment in the robot production line is required, and no negative impact on the cell production rate is generated.

However, those efficient design and programming methods must be based on an accurate industrial robot energy models to reduce EC [29]. To establish the EC model of industrial robots, it is necessary to calculate the dynamic characteristics of the manipulator. The dynamic modeling method is usually constructed by the famous Newton–Euler method or Lagrange formula, which requires the dynamic parameters of each link, such as mass, center of gravity, inertia tensor and friction coefficient [34]. In addition to the manipulator, the power flow modeling of permanent magnet synchronous motor (PMSM) and electric drive system is also an indispensable part of EC calculation. Therefore, the electrical parameters such as armature inductance, back emf constant, motor torque constant and rotor inertia of PMSM must be known [31]. Furthermore, the conduction and switching loss coefficient of the rectifier and inverter in the electrical drive system must also be known [11]. However, due to the commercial properties of industrial robots, the above robot dynamic parameters and electrical parameters will not be provided. Therefore, in engineering applications, the above parameters must be identified through complex parameter experimental identification [35], which makes the model-based optimization method challenging to apply in engineering and has considerable deviations [36].

With the development of data science and artificial intelligence, data-driven energy-saving methods are gradually applied in manufacturing [37–39]. He et al. proposed a data-driven method to reduce EC and identify sensitive features of EC from the original mechanical data [37]. Yan et al. adopted BP neural network to describe the mapping between maximum speed, maximum acceleration, starting point coordinates, end point coordinates, operation model and the EC of robot. [38]. To solve the limitations of path range and modeling efficiency of the method in [38], Yan et al. proposed an EC modeling strategy based on transfer learning to improve the modeling efficiency and model accuracy in the case of small sample size [39]. However, those researches focus on the linear and point-to-point (PTP) move modes, and influence of the load are not considered. Furthermore, the dynamic characteristic of the electromechanical system of the robot attributed from capacitive and inductive components in the drive system is ignored.

In light of those points, this paper focuses on reducing the EC of industrial robot based data-driven method. The contributions of this paper are threefold. First, the cause-and-effect relationship between motion variables and EC are analyzed with the consideration of the influence of capacitive and inductive components in the drive system. Second, the EC prediction model of time scaled trajectory based on long short-term memory neural network is proposed. Finally, the time scaling function is optimized based on the adaptive genetic algorithm (AGA) to reduce the EC of industrial robot.

2. Energy consumption analysis

2.1. Energy consumption analysis of industrial robot

In general, a typical industrial robot is usually driven by n PMSMs. The joint torque vector of the industrial robot, denoted as $\boldsymbol{\tau} \in \mathbb{R}^n$, can be obtained by a dynamic equation according to [20,40,41,33], which can be formulated as:

$$\boldsymbol{\tau} = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) + \boldsymbol{f}(\boldsymbol{q},\dot{\boldsymbol{q}}) + \boldsymbol{\tau}_{\mathrm{e}}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}}), \tag{1}$$

where $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ is the symmetric joint-space inertia matrix of the robot, $\boldsymbol{q} \in \mathbb{R}^n$, $\dot{\boldsymbol{q}} \in \mathbb{R}^n$ and $\ddot{\boldsymbol{q}} \in \mathbb{R}^n$ represent position, velocity and acceleration vectors of joint, respectively. $\boldsymbol{C} \in \mathbb{R}^{n\times n}$ is adopted to describe the Coriolis and centrifugal torque, $\boldsymbol{f} \in \mathbb{R}^n$ represents the equivalent friction torque on the joint, which mainly consists of the Coulomb friction and viscous friction, and it can be described as a function of $\boldsymbol{q},\dot{\boldsymbol{q}}$. $\boldsymbol{G} \in \mathbb{R}^n$ represents the torque due to gravity. $\boldsymbol{\tau}_{\mathrm{e}}$ represent the equivalent external torque on the joints, which is related to joint position, velocity, acceleration, due to the transformation relationship between the operation space and the joint space.

In Formula (1), the dynamic parameters such as mass, center of gravity, inertia tensor and friction coefficient of the robot affect the EC by affecting the joint torque. The effects of all dynamic parameters are encapsulated in the $\boldsymbol{M},\boldsymbol{C},\boldsymbol{G}$ and $\boldsymbol{f}$ matrices. The dynamic parameters and the four matrices are one-to-one mappings on the condition that $\boldsymbol{q},\dot{\boldsymbol{q}}$ and $\ddot{\boldsymbol{q}}$ are determined. Assume that the dynamic parameters in Eq. (1) are known, the power delivered to the mechanical system of the n-DOF serial robot is a function of $\boldsymbol{q},\dot{\boldsymbol{q}}$ and $\ddot{\boldsymbol{q}}$, denoted as φ_{mec}, ignoring efficiency of mechanical transmission, which can be formulated as:

$$P_{\mathrm{mec}} = \boldsymbol{\tau}^T\dot{\boldsymbol{q}} = \varphi_{\mathrm{mec}}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}}). \tag{2}$$

For PMSM, its dynamics and power input are complex and nonlinear, and are also affected by various disturbances and uncertainties [42]. Simply, the n PMSMs power input relative to the rotor coordinate system can be expressed as [29]:

$$P_{\mathrm{pmsm}} = P_{\mathrm{mec}} + P_{\mathrm{Cu}} + P_{\mathrm{Fe}} + P_{\mathrm{add}}, \tag{3}$$

where P_{add} represents miscellaneous losses and possible calculation error [10], which related to the absolute value of P_{pmsm} [29], and it can be formulated as:

$$P_{\mathrm{add}} = \varphi_{\mathrm{add}}(|P_{\mathrm{pmsm}}|). \tag{4}$$

And P_{Cu} and P_{Fe} represent copper loss and iron loss, respectively. While copper loss and iron loss are related to the motor torque and motor speed. Considering the linear relation between the joint torque, joint velocity, motor torque and motor speed [29], the copper loss and iron loss can be formulated as:

$$P_{\mathrm{Cu}} = \varphi_{\mathrm{Cu}}(\boldsymbol{\tau}\odot\boldsymbol{\tau}), \tag{5}$$

$$P_{\mathrm{Fe}} = \varphi_{\mathrm{Fe}}(|\dot{\boldsymbol{q}}|), \tag{6}$$

where $\odot$ denotes Hadamard product of two vectors.

Combined with Eq. (1)–(6), P_{pmsm} can be formulated as a function of $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$ and $\ddot{\boldsymbol{q}}$, and it can be written as:

$$P_{\mathrm{pmsm}} = \varphi_{\mathrm{pmsm}}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}}). \tag{7}$$

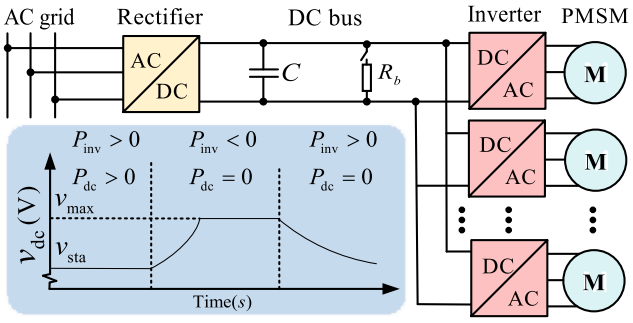


Fig. 1. Electric drive system.

The above expressions ignore the complex dynamics of input current and voltage of PMSM, but the calculation of power terms such as P_{Cu} and P_{Fe} are closely related to them, however, the difference between the two calculation is small, which can be neglected [29]. Furthermore, during the robot's movement, the input current and voltage of PMSM are affected by the servo controller, which uses the finite difference of input current and voltage to approach the differential [21]. Since the PMSM input current and voltage depend on the motion state of the robot, the calculation of P_{pmsm} depends not only on the position, velocity and acceleration of joints at the moment but also the values during the previous period. Namely, there exists a cause-and-effect function between P_{pmsm} and $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\ddot{\boldsymbol{q}}$.

The electric drive system composes of the rectifier, DC bus, and inverter, shown in Fig. 1. The rectifier extracts power from the industrial grid, and convert the AC current into DC current while the inverter adjusts frequency of the AC voltage based on pulse width modulation wave (PWM) according to the command of the motor servos. Since the power flow cannot reverse through the rectifier, the total power $P_{ac} \geq 0$.

To improve energy utilization, this drive system usually allows PMSMs to exchange energy through the DC bus [43,31]. When multiple axes are in regenerative braking state at the same time, the regenerative braking energy flows from the PMSMs to the DC bus through the inverter, charging the capacitor, as a result, the bus voltage v_{dc} can continue to rise from the standard value v_{sta} to the maximum value v_{max}. When $v_{dc} > v_{max}$, the braking resistor works, and the excess regenerative energy will be converted into Joule heat to avoid the further rise in the bus voltage. Since industrial robots frequently accelerate and decelerate while performing tasks, the charging and discharging processes of the capacitor are noticeable. Accordingly, the charging and discharging processes of the capacitor need to be analyzed. Note that the DC bus power flow of the electric drive system can be expressed as [11]:

$$P_{dc} = P_{inv} + P_{cap} + P_{rd}, \tag{8}$$

$$P_{cap} = \begin{cases} -P_{inv}, & \text{if } v_{sta} \leq v_{dc} \leq v_{max} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

$$P_{rd} = \begin{cases} -P_{inv}, & \text{if } v_{dc} > v_{max} \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where P_{dc} is the power input to the DC bus, P_{cap} is the power flowing into the DC bus capacitance, P_{rd} is the power dissipated through the brake resistance, P_{inv} is the power required by the inverters, which can be formulated as [11]:

$$P_{inv} = P_{pmsm} + P_{invlos}, \tag{11}$$

where P_{invlos} is the loss on the invert, which can be expressed as [11]:

$$P_{invlos} = \boldsymbol{i}_{pmsm}^T \boldsymbol{K}_{invc} \boldsymbol{i}_{pmsm} + \boldsymbol{k}_{invsw} \left| \boldsymbol{i}_{pmsm} \right|, \tag{12}$$

where the diagonal matrix $\boldsymbol{K}_{invc} \in \mathbb{R}^{n \times n}$ represents the conduction coefficients, while $\boldsymbol{k}_{invsw} \in \mathbb{R}^n$ is the switching losses coefficient. And $\boldsymbol{i}_{pmsm} \in \mathbb{R}^n$ is the PMSMs equivalent current, which can be formulated as [1]:

$$\boldsymbol{i}_{pmsm} = \boldsymbol{K}_{gea}^{-1} \boldsymbol{K}_{tor}^{-1} \boldsymbol{\tau}, \tag{13}$$

where $\boldsymbol{K}_{gea} \in \mathbb{R}^{n \times n}$ and $\boldsymbol{K}_{tor} \in \mathbb{R}^{n \times n}$ represent gear ratios and motor torque constants, respectively.

Based on the Kirchhoff's law, the DC bus exists the following differential equation:

$$C \frac{dv_{dc}}{dt} = \frac{P_{dc} - P_{inv} - P_{rd}}{v_{dc}}. \tag{14}$$

According to (8) to (10), P_{rd} and P_{dc} are determined by v_{dc}. The differential Eq. (14) demonstrates the v_{dc} is effected by P_{inv}, namely, P_{dc} is also effected by P_{inv} while P_{inv} is effected by $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$ and $\ddot{\boldsymbol{q}}$. Correspondingly, the value of P_{dc} is correlated with the current values of $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\ddot{\boldsymbol{q}}$ and their previous values due to the differential Eq. (14).

Then, the total power P_{ac} can be obtained [11,30]:

$$P_{ac} = P_{dc} + P_{reclos} + P_{other}, \tag{15}$$

$$P_{reclos} = K_{recc}\left(\frac{P_{dc}}{v_{dc}}\right)^2 + K_{recsw}\left|\frac{P_{dc}}{v_{dc}}\right|, \tag{16}$$

where K_{recc} and K_{recsw} are the conduction and switching-losses coefficients of rectifier, P_{reclos} is the rectifier loss, which is similar to P_{invlos} [11]. Other energy loss P_{other} is mainly original from the cooling fans, the PC in the control cabinet and the control panel, which is generally deal as a constant [30].

Therefore, since the regenerative energy can be exchanged through the DC bus or stored in the bus capacitor, the calculation of the total power P_{ac} will not only be related to the position, velocity and acceleration of joints at the current moment but also the period before. Consequently, based on the formula in this section, P_{ac} can be explained by the pattern:

$$P_{ac} = \varphi_{ac}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}), \tag{17}$$

where φ_{ac} represents the nonlinear function.

Assumed the joint position, velocity and acceleration trajectories, $\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)$ and $\ddot{\boldsymbol{q}}(t)$ are preplanned, the total EC of the robot can be calculated according to the energy integral model:

$$E_{ac} = \int_0^{t_f} \varphi_{ac}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})\, dt, \tag{18}$$

where t_f represents the end motion time of robot.

Generally, many of the dynamic and electrical parameters of robots are confidential and cannot be changed. The EC can only be optimized by setting the joint position, velocity and acceleration trajectories while not changing the hardware.

2.2. Energy consumption of robot under time scaling function

Generally, industrial robot task is conducted based on a corresponding reference joint trajectory $\boldsymbol{q}_r(t_r)$, which is preplanned to avoid the obstacles and singular configurations, where t_r represents the time for the reference frame. In a manufacturing system, the motion of an industrial robot can be achieved by scaling the reference trajectory at higher or low velocity to satisfy the cycle time or dynamic constraints. For example, an industrial robot can conduct the preprogrammed motion file with a feed-rate override α, and the task would be finished by $1/\alpha$ execution time of the preplanned task, which provides a convenient way to schedule the motion of the robot without modification of preplanned paths in the joint space. Thus, the relation between the reference time frame and the scaled time frame can be formulated as:

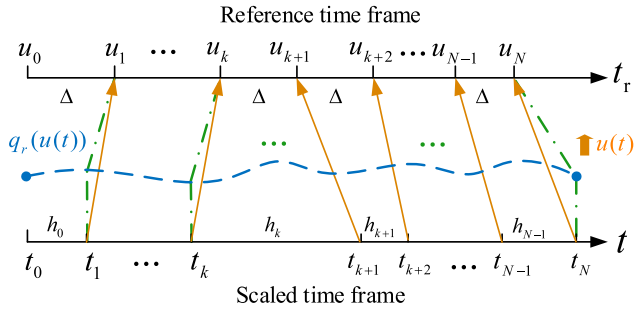


Fig. 2. Non-linear mapping between reference time frame and scaled time frame.

$$t_{\mathrm{r}} = u(t), \tag{19}$$

where $u(\cdot) \in \mathbb{R}^+$ is time scaling function, which is a monotonic increasing function.

It is worth noting that the time scaling function can be linear or nonlinear. And t represents the scaled time frame. Note that the scaled trajectory $\boldsymbol{q}_{\mathrm{s}}(t)$ actually equals to the reference trajectory $\boldsymbol{q}_{\mathrm{r}}(t_{\mathrm{r}})$ scaled in the time dimension, therefore, substituting t_{r} with $u(t)$ based on (19) can obtain the scaled trajectory, furthermore, the scaled velocity and acceleration trajectories can be obtained based on differentials, which can be formulated as:

$$\begin{aligned}
\boldsymbol{q}_{\mathrm{s}}(t) &= \boldsymbol{q}_{\mathrm{r}}(t_{\mathrm{r}}) = \boldsymbol{q}_{\mathrm{r}}(u(t)),\\
\dot{\boldsymbol{q}}_{\mathrm{s}}(t) &= \frac{d\boldsymbol{q}_{\mathrm{r}}(t_{\mathrm{r}})}{dt_{\mathrm{r}}}\dot{u}(t) = \dot{\boldsymbol{q}}_{\mathrm{r}}(t_{\mathrm{r}})\dot{u},\\
\ddot{\boldsymbol{q}}_{\mathrm{s}}(t) &= \frac{d^2\boldsymbol{q}_{\mathrm{r}}(t_{\mathrm{r}})}{dt_{\mathrm{r}}^2}\dot{u}^2 + \frac{d\boldsymbol{q}_{\mathrm{r}}(u)}{du}\ddot{u} = \ddot{\boldsymbol{q}}_{\mathrm{r}}(t_{\mathrm{r}})\dot{u}^2 + \dot{\boldsymbol{q}}_{\mathrm{r}}^2(t_{\mathrm{r}})\ddot{u}.
\end{aligned} \tag{20}$$

Assume the time scaling function $u(t)$ is second-order differentiable and the reference joint trajectory is known, the scaled joint trajectory is determined, namely, the scaled joint position, velocity and acceleration trajectories can be considered as functional of the reference joint position, velocity, acceleration trajectories and scaling function.

In practice, the industrial robot operates based on computer control, therefore, the joint position, velocity, acceleration reference trajectories and time scaling function are uniformly sampled with respect to the reference time frame. The time sequence is denoted as $\{t_{\mathrm{r},0}, t_{\mathrm{r},1}, \dots, t_{\mathrm{r},N}\}$. The sampling interval at the reference time frame is denoted as $\Delta = t_{\mathrm{r},k+1} - t_{\mathrm{r},k}$, which keeps constant. And the values of the reference joint position, velocity, acceleration and the time scaling function at the moment $t_{\mathrm{r}} = t_{\mathrm{r},k}$, $k \in \{0, 1, \dots, N\}$, are denoted as $\boldsymbol{q}_{\mathrm{r},k}$, $\dot{\boldsymbol{q}}_{\mathrm{r},k}$, $\ddot{\boldsymbol{q}}_{\mathrm{r},k}$ and u_k, respectively.

The corresponding time sequence with respect to the scaled time frame is denoted as $\{t_0, t_1, \dots, t_N\}$. It is worth noting that the sampling interval $h_k = t_{k+1} - t_k$ in the scaled time frame is non-uniform if the time scaling function is nonlinear, which is demonstrated in Fig. 2. The corresponding vectors of the scaled joint position, velocity and acceleration at the moment $t = t_k$ are denoted as $\boldsymbol{q}_{\mathrm{s},k}$, $\dot{\boldsymbol{q}}_{\mathrm{s},k}$, $\ddot{\boldsymbol{q}}_{\mathrm{s},k}$. Therefore, the relation between joint motion vectors in the reference frame and in the scaled frame (20) can be reformulated in discrete formation as:

$$\begin{aligned}
\boldsymbol{q}_{\mathrm{s},k} &= \boldsymbol{q}_{\mathrm{r},k},\\
\dot{\boldsymbol{q}}_{\mathrm{s},k} &= \dot{\boldsymbol{q}}_{\mathrm{r},k} \cdot \dot{u}_k,\\
\ddot{\boldsymbol{q}}_{\mathrm{s},k} &= \ddot{\boldsymbol{q}}_{\mathrm{r},k} \cdot \dot{u}_k^2 + \dot{\boldsymbol{q}}_{\mathrm{r},k} \cdot \ddot{u}_k.
\end{aligned} \tag{21}$$

Assume the mapping between $\boldsymbol{q}_{\mathrm{s},k}$ and $\boldsymbol{q}_{\mathrm{r},k}$ is known, the EC of an industrial robot tracks a scaled joint trajectory can be calculated as:

$$E_{\mathrm{ac}} = \sum_{k=0}^{N-1} \varphi_{\mathrm{ac}}(\boldsymbol{q}_{\mathrm{s},k}, \dot{\boldsymbol{q}}_{\mathrm{s},k}, \ddot{\boldsymbol{q}}_{\mathrm{s},k}) h_k. \tag{22}$$

3. Energy consumption prediction based on LSTM

According to the EC analysis, the industrial robot has a complex energy characteristic, and it is difficult to calculate the EC based the energy analyzed model of industrial robot for two reasons: First, differential equations describing the current or voltage of the PMSM and DC-bus capacitor are difficult to solve, therefore, it is challenging to find an explicit and precise analytical EC model. To simplify the model, inductance and capacitors in the driven system are often neglected [31,10,29]. Second, the EC model includes lots of dynamic parameters, electrical parameters, and control parameters, which cannot be obtained in practice.

The deep learning model is a deep neural network (NN) model with multiple nonlinear mapping layers. It can abstract the input signal layer by layer and extract features to mine more profound potential rules [39]. Therefore, a depth neural network is adopted to describe the relationship between the EC and the time scaling function for the specific reference trajectory in this paper.

In addition, due to Eq. (7) and (14), sequences of variables, such as the current of the dq-axis and DC-bus voltage, are the cause-and-effect sequences. Namely, the EC of industrial robots is not exclusively associated with the current joint position, velocity, and acceleration, but rather, it is also influenced by the robot's joint position, velocity, and acceleration at the preceding moment.

Note that, recurrent neural network (RNN) introduces the concept of time series into the network structure design, making it more adaptable in time series data analysis. Furthermore, the long short-term memory neural network makes up for the problems of gradient disappearance, gradient burst, and insufficient long-term memory ability of RNN so that the LSTM can effectively use long-distance time information [44], which is suitable for solving time-dependent cause-and-effect sequence prediction problems. Therefore, LSTM neural network is adopted to predict the EC of industrial robot tracking to the preplanned path under different time scaling functions.

LSTM neural network is a recurrent neural network with LSTM cells as hidden layers. The LSTM cell is the basic unit of the LSTM neural network, and the cell schematic is shown in Fig. 3.

The LSTM cell has three gate structures, forget gate, input gate, and output gate, all of which are one-layer fully connected neural networks. LSTM cells selectively transmit information through the gate structure to achieve the purpose of information memory, extraction and prediction. Therefore, its forward calculation principle can be expressed as [45]:

$$\begin{aligned}
\boldsymbol{f}_{t(k)} &= \sigma\left(\boldsymbol{W}_{\mathrm{xf}}\boldsymbol{x}_{t(k)} + \boldsymbol{W}_{\mathrm{hf}}\boldsymbol{h}_{t(k-1)} + \boldsymbol{b}_{\mathrm{f}}\right),\\
\boldsymbol{i}_{t(k)} &= \sigma\left(\boldsymbol{W}_{\mathrm{xi}}\boldsymbol{x}_{t(k)} + \boldsymbol{W}_{\mathrm{hi}}\boldsymbol{h}_{t(k-1)} + \boldsymbol{b}_{\mathrm{i}}\right),\\
\boldsymbol{c}_{t(k)} &= \boldsymbol{i}_{t(k)} \odot \tanh\left(\boldsymbol{W}_{\mathrm{xc}}\boldsymbol{x}_{t(k)} + \boldsymbol{W}_{\mathrm{hc}}\boldsymbol{h}_{t(k-1)} + \boldsymbol{b}_{\mathrm{c}}\right)\\
&\quad + \boldsymbol{c}_{t(k-1)} \odot \boldsymbol{f}_{t(k)},\\
\boldsymbol{o}_{t(k)} &= \sigma\left(\boldsymbol{W}_{\mathrm{xo}}\boldsymbol{x}_{t(k)} + \boldsymbol{W}_{\mathrm{ho}}\boldsymbol{h}_{t(k-1)} + \boldsymbol{b}_{\mathrm{o}}\right),\\
\boldsymbol{h}_{t(k)} &= \boldsymbol{o}_{t(k)} \odot \tanh\left(\boldsymbol{c}_{t(k)}\right),
\end{aligned} \tag{23}$$

where $\boldsymbol{f}_{t(k)}$, $\boldsymbol{i}_{t(k)}$ and $\boldsymbol{o}_{t(k)}$ are the output of forget gate, input gate and output gate at time t_k, respectively. $\boldsymbol{W}_{\mathrm{xf}}$, $\boldsymbol{W}_{\mathrm{hf}}$, $\boldsymbol{W}_{\mathrm{xi}}$, $\boldsymbol{W}_{\mathrm{hi}}$, $\boldsymbol{W}_{\mathrm{xc}}$, $\boldsymbol{W}_{\mathrm{hc}}$, $\boldsymbol{W}_{\mathrm{xo}}$ and $\boldsymbol{W}_{\mathrm{ho}}$ are the weight coefficient matrix corresponding to each gate, respectively. $\boldsymbol{b}_{\mathrm{f}}$, $\boldsymbol{b}_{\mathrm{i}}$, $\boldsymbol{b}_{\mathrm{c}}$ and $\boldsymbol{b}_{\mathrm{o}}$ are bias term corresponding to each gate, respectively. $\boldsymbol{h}_{t(k)}$ and $\boldsymbol{c}_{t(k)}$ are the output and state of the cell at time t_k, σ and tanh are sigmoid activation function and hyperbolic tangent activation function, which are defined as:

$$\begin{aligned}
\sigma(z) &= \frac{1}{1+e^{-z}},\\
\tanh(z) &= \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}.
\end{aligned} \tag{24}$$

To deeply investigate the deep potential rules between the total power and the input trajectory variables and extract the features of the discrete input variables set $\{\dot{u}_k, \ddot{u}_k, \boldsymbol{q}_{\mathrm{r},k}, \dot{\boldsymbol{q}}_{\mathrm{r},k}, \ddot{\boldsymbol{q}}_{\mathrm{r},k}\}$, $k \in \{0, 1, \dots, N\}$, the

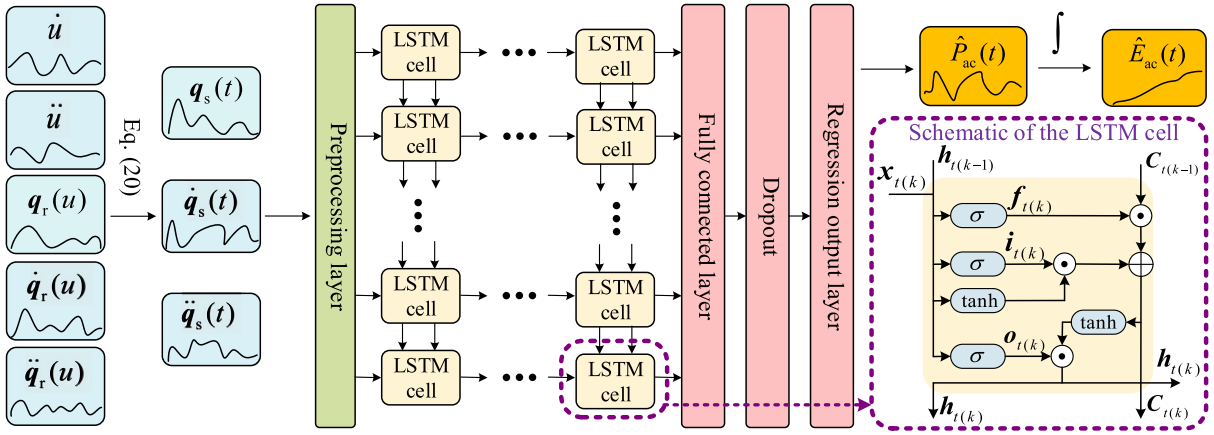


Fig. 3. Schematic of the LSTM cell and the proposed neural network.

Table 1
Input and output variables of the proposed neural network.

Variable name	Value
Input of the proposed neural network	$\{\dot{u}_k, \ddot{u}_k, \boldsymbol{q}_{\mathrm{r},k}, \dot{\boldsymbol{q}}_{\mathrm{r},k}, \ddot{\boldsymbol{q}}_{\mathrm{r},k}\}$
Input of the preprocessing layer	$\{\boldsymbol{q}_{\mathrm{s},k}, \dot{\boldsymbol{q}}_{\mathrm{s},k}, \ddot{\boldsymbol{q}}_{\mathrm{s},k}\}$
Output of the regression output layer	$\hat{P}_{\mathrm{ac}}(t_k)$
Output of the proposed neural network	$\hat{E}_{\mathrm{ac}}$

LSTM-based deep learning neural network adopted for EC calculation is shown in Fig. 3. The corresponding input and output variables are shown in Table 1.

The proposed neural network includes six functional modules: trajectory scaling, data preprocessing layer, stacking LSTM layer, full connection layer, dropout layer and regression output layer. Trajectory scaling module directly generates the scaled robot joint trajectory in time domain based on the reference joint trajectory and the scaling function. The data preprocessing layer standardizes the input data and extracts a certain proportion of data as the test set by stratified sampling. The remaining data are divided into the training set and validation set by the K-fold cross-validation method to train the proposed neural network. The stacked LSTM layer mines the deep rules between the total power and the input variables. The fully connected layer is used to reduce the dimension of stacked LSTM output data, and the dropout layer is added to prevent overfitting. The regression output layer is used to output and denormalize the total power curve.

The weight parameters of proposed neural network are initialized using the Glorot initializer, the forget gate bias is initialized to 1, and the remaining bias is initialized to 0. The training algorithm uses the back propagation algorithm. To accurately predict the EC of the whole operation process of the robot, mean square error (MSE) and mean absolute error (MAE) are used as the loss function to calculate the loss of the total power curves. And the mean absolute percentage error (MAPE) is used to characterize the prediction error of the proposed neural network for total EC. Those metrics are calculated as follows:

$$\mathrm{MSE} = \frac{1}{L \cdot N} \sum_{i=1}^{L} \sum_{k=0}^{N} (P_{\mathrm{ac},i}(t_k) - \hat{P}_{\mathrm{ac},i}(t_k))^2, \tag{25}$$

$$\mathrm{MAE} = \frac{1}{L \cdot N} \sum_{i=1}^{L} \sum_{k=0}^{N} |P_{\mathrm{ac},i}(t_k) - \hat{P}_{\mathrm{ac},i}(t_k)|, \tag{26}$$

$$\mathrm{MAPE} = \frac{100\%}{L} \sum_{i=1}^{L} \frac{\left|E_{\mathrm{ac},i} - \hat{E}_{\mathrm{ac},i}\right|}{E_{\mathrm{ac},i}}, \tag{27}$$

where $\hat{P}_{\mathrm{ac},i}(t_k)$ is the output prediction by the proposed neural network for the input variable set of the ith sample. $E_{\mathrm{ac},i}$ and $\hat{E}_{\mathrm{ac},i}$ are the actual and predicted values of the total EC of the ith scaling trajectory, respectively. L is the total number of samples used to evaluate the prediction effect. Other detailed network training parameters, including learning rate, hidden layer dimension, batch size, *etc.*, will be discussed in the case study section.

Once the proposed neural network has been trained, the power sequence $\hat{P}_{\mathrm{ac}}(t_k)$ for any scaled trajectory can be obtained by the proposed neural network, $\hat{P}_{\mathrm{ac}}(t_k)$ can be integrated to calculate the total EC, which is the cost to minimize later:

$$\hat{E}_{\mathrm{ac}} = \sum_{k=0}^{N-1} \hat{P}_{\mathrm{ac}}(t_k) h_k. \tag{28}$$

Remark 1. The proposed neural network can predict the EC of an industrial robot tracking any scaled trajectory, and all the scaled trajectories originate from the same reference trajectory. Therefore, the samples in the training and test set must be about the same reference trajectory sequences. According to the EC analysis, the dynamic parameters of industrial robot, the parameters of the PMSMs and the parameters of the reference trajectory are invariant, correspondingly, all the samples in the training and test set should be obtained based on the same robot model with the same load.

Remark 2. The trajectory scaling module is designed to conduct the scaling operation for two reasons: First, the scaling operation can be conduct based on explicit equation, which does not need be fitted based on neural network; Second, it can reduce the dimension of the input of the LSTM neural network, which can reduce the training sample quantity and simplify the LSTM neural network.

4. Energy optimization based on AGA

Trajectory planning is an ecologically efficient EC reduction method [29]. According to the analysis in Section 2.2, solving the energy-optimal trajectory planning problem is to find a scaling function $u(t)$, which minimizes the EC function (28). Since the reference trajectory has been uniformly divided by $N+1$ sampling points in Section 2.2. Assume the sampling frequency is high enough, and the acceleration of u can consider as a constant.

Note that $\dot{u}$ and $\ddot{u}$ are the first order and the second order derivatives of u with respect to t, respectively. Since $\ddot{u}$ can be regarded as a constant during $[t_k,\ t_{k+1}]$, the state of u can be formulated as:

$$\begin{bmatrix} u_{k+1} \\ \dot{u}_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & h_k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_k \\ \dot{u}_k \end{bmatrix} + \begin{bmatrix} \frac{h_k^2}{2} \\ h_k \end{bmatrix} \ddot{u}_k. \tag{29}$$

Note that the value of u_k is known, namely, $u_k = k \cdot \Delta$, because the sample periods in the reference time frame are uniform. However, the corresponding sample periods of the scaled trajectory are non-uniform,

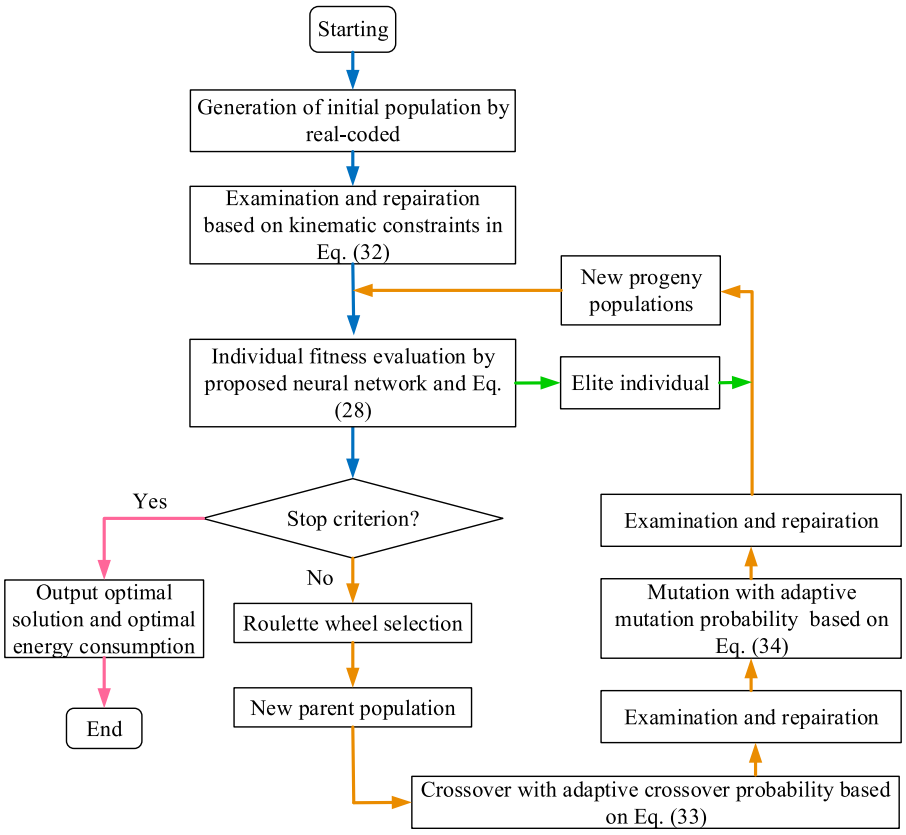


Fig. 4. Flowchart of AGA.

therefore, Eq. (29) cannot determine the value of t when the state variable $[u_k,\ \dot{u}_k]^T$ is known.

During the period $[t_k,\ t_{k+1}]$, u increases with constant acceleration, there exists $\Delta = h_k\dot{u}_k + h_k^2\ddot{u}_k/2$, the following equation holds:

$$h_k\ddot{u}_k = \frac{2\Delta}{h_k} - 2\dot{u}_k. \tag{30}$$

Select $[t_k\ \dot{u}_k]^T$ as the state variable. Based on $t_{k+1} = t_k + h_k$, Eq. (30) and (29), a new state equation can be written as:

$$\begin{bmatrix} t_{k+1} \\ \dot{u}_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} t_k \\ \dot{u}_k \end{bmatrix} + \begin{bmatrix} h_k \\ \frac{2\Delta}{h_k}, \end{bmatrix}. \tag{31}$$

Assume state variables are given, t_k and $\dot{u}_k$ are directly known, u_k can be obtained according to the index, and $\ddot{u}_k$ can be obtained by the difference operation based on the adjacent state variable. Therefore, the mapping between the reference time frame and the scaled time frame can be determined, and the reference trajectory can be scaled to establish a complex relationship between variables set $\{\dot{u}_k, \ddot{u}_k, \boldsymbol{q}_{r,k}, \dot{\boldsymbol{q}}_{r,k}, \ddot{\boldsymbol{q}}_{r,k}\}$ and total EC through the proposed neural network.

Since the nonlinear function between total EC and time scaling trajectory variables is established by proposed neural network, the total EC optimization problem can be formulated as:

$$\begin{aligned} \text{Min}\quad & \hat{E}_{ac} \\ \text{s. t.}\quad & -\dot{\boldsymbol{q}}_{max} \le \dot{\boldsymbol{q}}_{s,k} \le \dot{\boldsymbol{q}}_{max}, \\ & -\ddot{\boldsymbol{q}}_{max} \le \ddot{\boldsymbol{q}}_{s,k} \le \ddot{\boldsymbol{q}}_{max}, \\ & t_N \le t_{max}, \end{aligned} \tag{32}$$

where $\dot{\boldsymbol{q}}_{max}$, $\ddot{\boldsymbol{q}}_{max}$, and t_{max} are the thresholds of joint velocity, acceleration and execution time. This problem is a typical combinatorial optimization problem, which has been proved to be NP-hard.

Genetic algorithm has global optimization ability, strong adaptability, strong robustness, which is suitable for solving NP-hard problems [46]. Thus, this study uses an adaptive genetic algorithm with an elite retention strategy to overcome the drawbacks of classic genetic algorithms, which is prone to falling into local optimums. In the elite retention approach, the individual with the highest fitness value is immediately chosen to replace the poorest one in the next progeny population. This strategy can prevent the loss of beneficial genes caused by crossover and mutation [30]. The flow chart of AGA optimization EC is shown in Fig. 4.

Based on Eq. (31), this study selects $\dot{u}_k$ sequence as the decision variables and the proposed neural network as the fitness function to optimize the EC of the industrial robot. The chromosome is encoded in real numbers and contains $N+1$ genes, which describes the $\dot{u}$ value at each sampling point of the robot scaling trajectory.

To retain excellent individuals and avoid falling into local optimization, the probability of crossover and mutation is adaptively modified according to the fitness of the individual in the current generation. By introducing the sigmoid function, the adaptive crossover and mutation probabilities can be formulated as [30]:

$$P_c = \begin{cases} P_{cmax} & , fit_c < fit_{avg} \\ P_{cmin} + \dfrac{P_{cmax} - P_{cmin}}{1 + \exp\left[\dfrac{A\left(fit_c - fit_{avg}\right)}{fit_{max} - fit_{avg}}\right]} & , fit_c \ge fit_{avg} \end{cases} \tag{33}$$

$$P_m = \begin{cases} P_{mmax} & , fit_i < fit_{avg} \\ P_{mmin} + \dfrac{P_{mmax} - P_{mmin}}{1 + \exp\left[\dfrac{A\left(fit_c - fit_{avg}\right)}{fit_{max} - fit_{avg}}\right]} & , fit_i \ge fit_{avg} \end{cases} \tag{34}$$

where P_{cmax} and P_{cmin} are the crossover probability's upper and lower bounds, respectively, while P_c is crossover probability for individual. P_m stands for the mutation probability, while P_{mmax} and P_{mmin} are its upper and lower bounds, respectively. And fit_i represents an individual's fitness level, fit_{max} and fit_{avg} stand for the population's

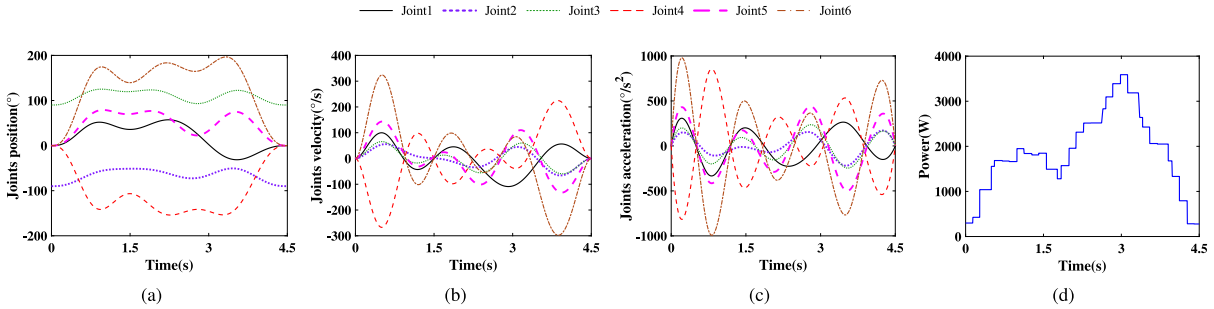


Fig. 5. Experimental results of the robot tracking the reference trajectory. (a) Joint position curve. (b) Joint velocity curve. (c) Joint acceleration curve. (d) Power curve.

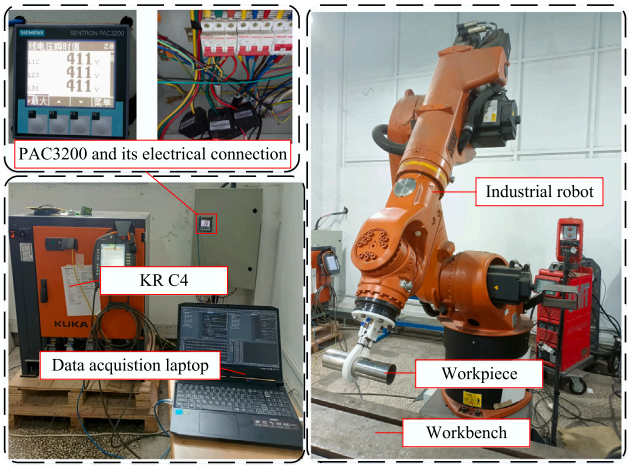


Fig. 6. The experiment setup.

Table 2
Restriction conditions of the robot's motion.

Constrains	Index	Value	Index	Value
Joint velocity (°/s)	Joint 1	128	Joint 4	260
	Joint 2	102	Joint 5	245
	Joint 3	128	Joint 6	322
Joint acceleration (°/s²)	1000			

maximum and average fitness level, respectively. fit_c is the average fitness level of two crossover individual. A is an adjustable hyperparameter. The adaptive crossover and mutation mechanism makes the crossover probability and mutation probability change dynamically according to the individual fitness, which increases the possibility of mutation of individuals with smaller fitness, and saves individuals with higher fitness to the next generation. Therefore, this mechanism can expand the overall search range in the early stage of evolution, and stabilize the population and preserve the excellent genes in the later stage of evolution.

5. Case study

5.1. Experimental environment

To validate the accuracy and efficacy of the proposed method, a series of pick-and-place experiments were conducted on a KR60-3 industrial robot made by KUKA Co., Ltd, shown in Fig. 6. The KR C4 controller is used to drive the robot. The robot was linked via Ethernet KRL to a laptop, which has an 11th Gen Intel® Core™ i5-11400H 6-Core CPU running at 2.7 GHz and 16 GB of RAM. Joint velocity and acceleration were calculated using the differential operation based on joint position sequence, and the joint position was sampled by the laptop at a sampling interval of 10 ms. Table 2 contains a list of restriction conditions of the robot's motion. The power of the robot was measured at 10 Hz by using the Sentron PAC3200 multipurpose measurement device, made by Siemens Co., Ltd.

Table 3
Parameters of proposed neural network.

Index	Parameters	Value
1	Input dimension	18
2	Number of hidden layer nodes	32
3	Layers of stacked networks	3
4	Dropout number of full connected layer	0.3
5	Maximum epoch	50
6	Number of batch size	5
7	Learning rate	$1 * 10^{-3}$
8	Optimizer	Adam

The reference trajectory of pick-and-place of the robot was generated by the robot teaching device according to the teaching point. Since robots usually work in full-speed mode in practice. The execution speed was set to 80% of the robot's maximum execution speed, which is a fundamental operational parameter of the teaching device. As seen in Fig. 5, the robot tracked the reference trajectory while the joint position, velocity, acceleration and power were sampled.

5.2. Accuracy validation of the prediction model

To precisely estimate the EC of the robot using the proposed neural network model, the number $N + 1$ of sampling points of reference trajectory is set to 226. The first derivative of the time scaling function was set to $0 < \dot{u} < 1.25$, which is limited by the maximum speed of the robot. Multiple groups of $\dot{u}$ sequences are randomly selected, each sequence contains $N+1$ randomly generated points of $\dot{u}$. Then, the reference trajectory was scaled by combining Eq. (20) and state Eq. (31). After screening, 500 groups of scaling trajectories that satisfying the joint velocity and acceleration constraints in Eq. (32) are obtained. The execution time threshold t_{max} was set to 20 s in Eq. (32). The robot tracked the scaled trajectory while the joint position, velocity, acceleration and power were sampled. And 500 sets of scaled joint position, velocity, acceleration, and power data were standardized for fast convergence. The proposed neural network hyperparameters were set as the parameters listed in the Table 3. The stratified sampling method extracted 20% of the samples from the standardized data set as the test set. The remaining data was divided into a training set and a validation set using a 5-fold cross-validation method. After the training was completed, the prediction effect of the five models was tested using the test set data, and the results are shown in Table 4. Table 4 shows that after the proposed neural network was trained under different cross-validation data partitions, the MSE on the test set was less than 2473.14 W^2, the MAE was less than 48.13 W, and the MAPE was less than 4.21%.

To verify the effectiveness of the proposed neural network for the prediction of cause-and-effect sequence, this section replaces the LSTM

Table 4
Comparison of prediction results of different methods on test set.

Method	Evaluation metric	5-fold cross-validation					Average
		1	2	3	4	5	
Proposed NN	MSE (W^2)	1711.54	2220.18	1833.44	2035.74	2548.14	2069.81
	MAE (W)	39.43	44.94	40.84	43.12	48.13	43.29
	MAPE (%)	4.02	4.17	4.04	4.13	4.21	4.11
RNN based NN	MSE (W^2)	5572.84	6979.05	5997.24	6519.85	7974.93	6608.78
	MAE (W)	71.18	79.58	73.79	76.96	85.09	77.32
	MAPE (%)	4.87	5.05	4.94	4.97	5.19	5.00
GRU based NN	MSE (W^2)	3578.38	4775.23	4017.76	4218.17	5352.34	4388.38
	MAE (W)	56.74	65.42	60.31	61.87	69.38	62.74
	MAPE (%)	4.46	4.66	4.53	4.60	4.72	4.59
ECPE	MSE (W^2)						3987.98
	MAE (W)			–			59.88
	MAPE (%)						4.52

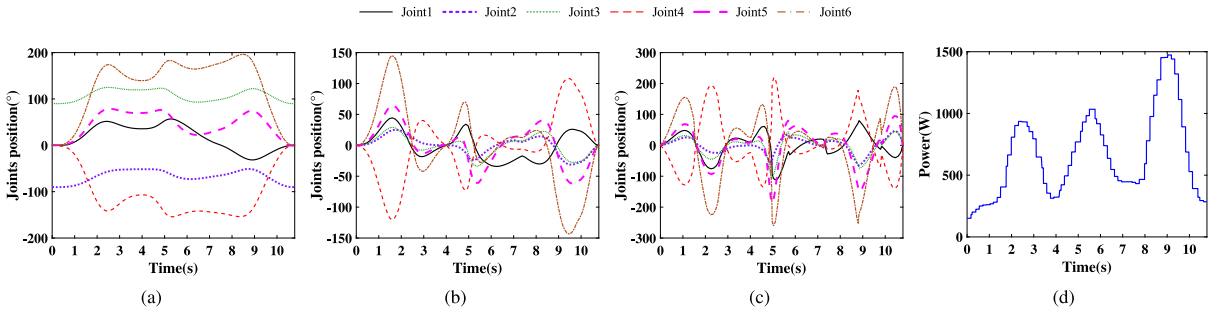


Fig. 7. Experimental results of the robot tracking the optimal trajectory based on AGA. (a) Joint position curve. (b) Joint velocity curve. (c) Joint acceleration curve. (d) Power curve.

cell with the basic RNN cell or the gated recurrent cell that simplifies the gating mechanism of the LSTM for comparison. The internal mechanisms of the basic RNN cell and the GRU cell can be expressed as Eqs. (35) and (36), respectively, as follows:

$$\boldsymbol{h}_{t(k)} = \tanh\left(\boldsymbol{W}_{\mathrm{xh}}\boldsymbol{x}_{t(k)} + \boldsymbol{W}_{\mathrm{hh}}\boldsymbol{h}_{t(k-1)} + \boldsymbol{b}_{\mathrm{h}}\right), \tag{35}$$

$$\begin{aligned}
\boldsymbol{r}_{t(k)} &= \sigma\left(\boldsymbol{W}_{\mathrm{xr}}\boldsymbol{x}_{t(k)} + \boldsymbol{W}_{\mathrm{hr}}\boldsymbol{h}_{t(k-1)} + \boldsymbol{b}_{\mathrm{r}}\right),\\
\boldsymbol{z}_{t(k)} &= \sigma\left(\boldsymbol{W}_{\mathrm{xz}}\boldsymbol{x}_{t(k)} + \boldsymbol{W}_{\mathrm{hz}}\boldsymbol{h}_{t(k-1)} + \boldsymbol{b}_{\mathrm{z}}\right),\\
\tilde{\boldsymbol{h}}_{t(k)} &= \tanh\left(\boldsymbol{W}_{\mathrm{xh}}\boldsymbol{x}_{t(k)} + \boldsymbol{W}_{\mathrm{hrh}}(\boldsymbol{h}_{t(k-1)} \odot \boldsymbol{r}_{t(k)}) + \boldsymbol{b}_{\mathrm{h}}\right),\\
\boldsymbol{h}_{t(k)} &= \boldsymbol{z}_{t(k)} \odot \tilde{\boldsymbol{h}}_{t(k)} + (1 - \boldsymbol{z}_{t(k)}) \odot \boldsymbol{h}_{t(k-1)},
\end{aligned} \tag{36}$$

where $\boldsymbol{z}$ is the update gate, $\boldsymbol{r}$ is the reset gate, $\tilde{\boldsymbol{h}}$ is the candidate hidden state, and the other symbols have the same meaning as LSTM. The parameter initialization method and training parameters of the replaced network were the same as those of the proposed neural network. In addition, the energy consumption parameterization equation (ECPE) in [33] was also adopted for EC prediction comparison. Table 4 shows the comparison results of the above four models. MAE measures the average distance between the predicted value and the actual value, MSE measures the deviation and implied volatility between the predicted value and the actual value, and MAPE reflects the relative error of the total EC. The results show that the proposed neural network MAE, MSE and MAPE have smaller values and stronger generalization ability.

Compared with the neural network based on the basic RNN cell, the neural network based on the GRU cell and the ECPE method, the proposed prediction neural network has the highest prediction accuracy, and the neural network based on the basic RNN cell has the largest relative prediction error. This may be because, compared with the neural network based on LSTM, the neural network based on the basic RNN cell has relatively weak time-dependent sequence processing ability and cannot retain more input information and hidden state information. Additionally, The GRU-based neural network simplifies the gating mechanism of LSTM, with only update gates and reset gates, so the expression ability and fitting ability of the model may be limited.

Theoretically, the prediction based on the EC parameterization equation should have the highest accuracy among the four methods, however, the result of the parameterization equation based method is slightly larger than the proposed method. The reason is that the prediction accuracy of EC parameterization equation strongly depends on the accuracy of the parameters in the equation, those parameters are not provided by the robot manufacturer and can only be identified by experiments, which results in the prediction error.

5.3. Effectiveness validation of the optimization method

Several experiments were conducted to validate the effectiveness of the optimization method based on the proposed prediction method. The reference trajectory in Section 5.2 was optimized, and the adaptive genetic algorithm proposed in Section 4 was adopted to optimized discrete sequence of $\dot{u}_k$. Furthermore, to reduce the error, the average of the proposed neural network output values under different cross-validation data partition training in Table 4 was used as the predicted value of EC in the optimization process. The parameters of AGA are listed in Table 5, and the algorithm was coded by Python language. In Section 5.2, the execution time threshold $t_{\max}$ in the optimization objective Eq. (32) was set to 20 s. The initial population was generated using the random function $Numpy.random.uniform$ in the Numpy library of Python. The iteration ceased if the optimal answer across all participants keeps the same after 2000 iterations or the iteration step exceeds 10,000, the detail of the AGA algorithm execution process is shown in Fig. 4.

Under the above conditions, the search process is shown in Fig. 9(a). After 2901 iterations, the iteration stopped, and the optimized $\dot{u}_k$ sequence was obtained, and details are shown in Table 6. The predicted EC after optimized is 6744.84 J while the corresponding measured EC is 6706.78 J, and relative error is 0.57%. The corresponding partial chromosome of $\dot{u}_k$ is shown in Table 6. The time scaling function and the optimal scaling trajectory based on the optimized $\dot{u}_k$ chromosome are shown in Fig. 9(b) and 7, respectively. By integrating the measured

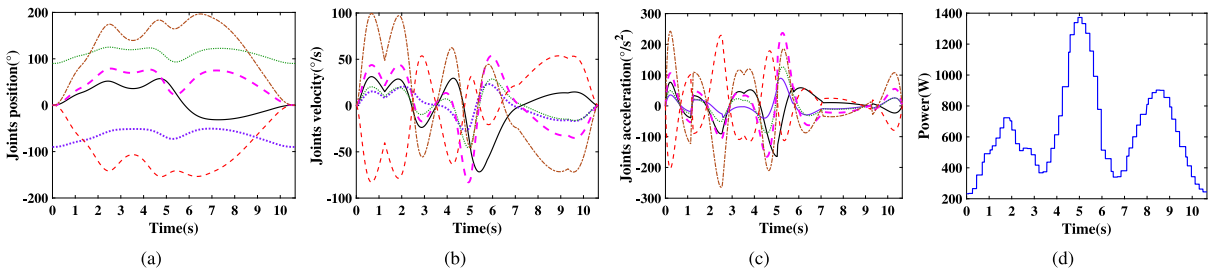


Fig. 8. Experimental results of the robot tracking the optimal trajectory based on DPBEPE. (a) Joint position curve. (b) Joint velocity curve. (c) Joint acceleration curve. (d) Power curve.

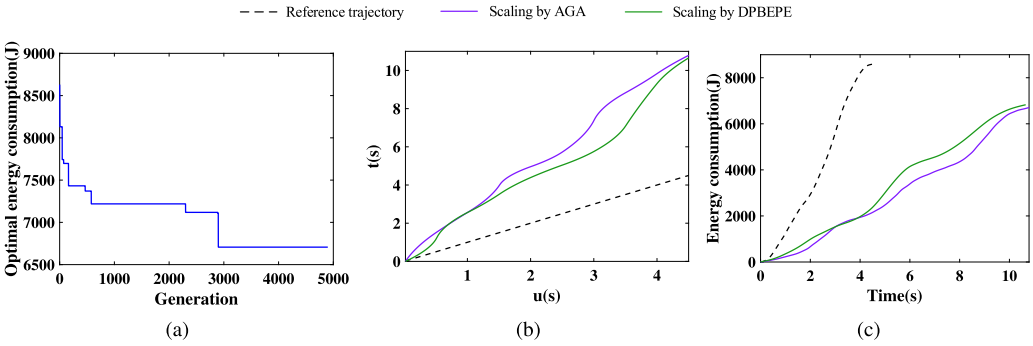


Fig. 9. Under t_{max} = 20 s restriction: (a) AGA search process. (b) Comparison of the time-scaling function obtained by different methods. (c) Comparison of the EC obtained by different methods.

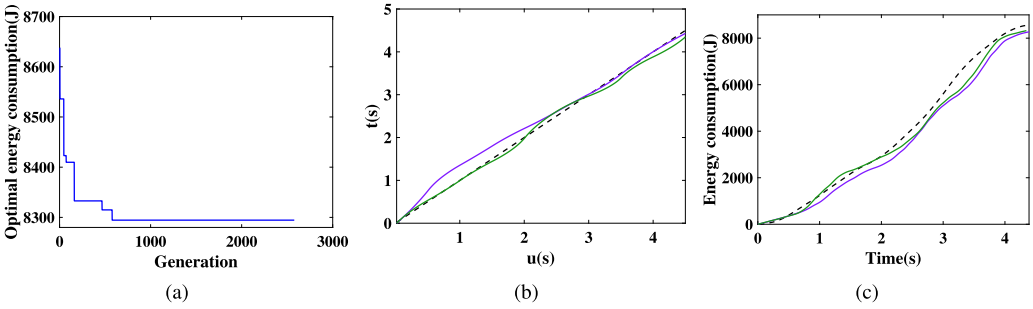


Fig. 10. Under t_{max} = 4.5 s restriction: (a) AGA search process. (b) Comparison of the time-scaling function obtained by different methods. (c) Comparison of the EC obtained by different methods.

Table 5
Parameters of AGA.

Index	Parameters	Value
1	Maximum generation	10000
2	Population size	200
3	Elite selection	10
4	Adjustable hyper-parameter A	1000
5	P_{cmax}	0.8
6	P_{cmin}	0.6
7	P_{mmax}	0.1
8	P_{mmin}	0.03

power curve in Fig. 7(d), the EC curve after optimized by AGA is shown in Fig. 9(c). According to Fig. 9(c), AGA reduced the EC from 8637.41 J to 6706.78 J, which is reduced by 22.35%.

For comparison, the dynamic programming based on energy parameterization equation (DPBEPE) in [33] was conducted. The dynamic programming (DP) grid parameters N_u, N_v and N_t in [33] was set as 225, 125 and 125, respectively. After optimized, the actual EC was reduced to 6827.20 J while the corresponding predicted EC was 6754.56 J, and relative error is 1.08%. Compared to the reference trajectory, EC is reduced by 20.96%. The corresponding $\dot{u}_k$ is shown in Table 6. And the time scaling function, optimal energy trajectory, and optimal EC curve are shown in Fig. 9(b), 9 (c), and 8, respectively.

Note that this paper argues that the time threshold t_{max} in the optimization objective Eq. (32) was considered to meet the constraints of the production cycle. However, to verify the advantages of the proposed method in reducing the EC of the given reference trajectory, the time threshold was set as the running time of the reference trajectory to search for the optimal scaling trajectory that reduces both EC and time. The corresponding experimental results are shown in Table 6 and Fig. 10. The results show that under the given time constraint, the proposed method and DPBEPE method reduce the EC by 5.39% and 4.64%, respectively.

The experimental results revealed that both the proposed and the DPBEPE methods in [33] can reduce the EC under the given constraint. Theoretically, the DPBEPE method can achieve a better result, because the method is based dynamic programming, which would search the whole solution space to obtain the global optimal solution. However, Figs. 9(b) and 9(c) demonstrate the proposed method can achieve a slightly better result. Two reasons attribute to this result: First, the DPBEPE predict the EC based on the energy characteristic parameterization equation, the accuracy of the corresponding EC prediction is less than the proposed LSTM based method, according to Section 5.2.

Table 6
Detailed optimization scheme.

Method	t_{max} (s)	Part chromosome of $\dot{u}_k$	Reference trajectory EC(J)	Optimal EC(J)	Measured EC(J)	Relative error (%)	Reduction of EC (%)
AGA	20	{0.200 0.216 0.230 … 0.584 0.589 0.595}	8637.41	6706.78	6706.78	0.57	22.35
DPBEPE		{0.650 0.638 0.625 … 0.477 0.485 0.492}		6754.56	6827.20	1.08	20.96
AGA	4.5	{0.567 0.593 0.618 … 0.752 0.723 0.693}	8637.41	8337.74	8171.85	2.03	5.39
DPBEPE		{0.650 0.643 0.637 … 1.093 1.100 1.107}		8466.43	8236.63	2.79	4.64

Second, the DPBEPE optimizes the EC by dynamic programming, which searches the optimal solution based on the discrete grid of the state, and the optimization result largely depend on the grid density. The higher the grid density is, the better optimized result it can obtain but the computation time increases exponentially, while the lower grid density cost less computation time at the costs of poorer optimized result.

6. Conclusion

Due to the wide distribution and low energy efficiency of industrial robots, the trajectory planning of the industrial robot has great energy-saving potential. This paper proposed a data-driven method for the prediction and optimization of the robot EC. Based on the influence of the capacitive and inductive components in the drive system, the cause-and-effect relationship between robot EC and joint motion variables, was qualitative analyzed. And a LSTM based deep neural network is developed to predict the EC based on the cause-and-effect relationship between robot EC and joint motion variables. The adaptive genetic algorithm is adopted to reduce the EC based on the optimal time-variant scaling function. Experimental results demonstrate that the prediction error of the proposed LSTM based neural network is less than 4.21%, and the AGA-based optimization method can reduce the reference trajectory EC by 22.35%.

The unique advantage of the proposed method over the conventional prediction techniques is that it encapsulates the influence of dynamic response in DC bus and PMSMs into the LSTM based neural network, and it can extract the complex and implicit EC relations in a more comprehensive manner, which helps in increasing the prediction accuracy. Furthermore, the proposed data-driven optimization method does not need any parameters of the industrial robot, which has a great advantage over the conventional model-based method in practice.

It is worth noting that the proposed LSTM based neural network is feasible for a given reference trajectory, once the reference trajectory changes, the neural network should be trained based on new sample data. Therefore, the proposed method is more suitable for EC of the mass production, the application in the small batch manufacturing might bring about numerous pre-experiments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge financial support from the National Key R&D Program of China (No. 2020AAA0109204), the National Science Foundation of China (No. 52075060, No. 51705050), the Natural Science Foundation of Chongqing, China (No. 2022NSCQ-MSX1629) and the Fundamental Research Funds for the Central Universities, China (No. 2022CDJKYJH024).

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